EXAMINATIONS COUNCIL OF ZAMBIA

Examination for General Certificate of Education Ordinary Level

# Additional Mathematics 4030/1

## Paper 1

**Tuesday** **15 AUGUST 2017**

**Additional Materials:**

Answer Booklet
Silent electronic calculator (non programmable)

**Time: 2 hours**

### Instructions to Candidates

Write your **name, centre number** and **candidate number** in the spaces on the Answer Booklet provided.

There are **12 questions** in this paper. Answer **all** questions.

Write your answers in the **Answer Booklet provided**.

If you use more than one Answer Booklet, **fasten** the Answer Booklets together.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

### Information for candidates

The number of marks is shown in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 80.

**The use of a non programmable electronic calculator is expected, where appropriate.**

**Cell phones are not allowed in the examination room.**

You are reminded of the need for clear presentation in your answers.

**Check the formulae overleaf.**

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## MATHEMATICS FORMULAE

### 1 ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

*Binomial Theorem*

$$(a+b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \ldots\ldots + \binom{n}{r} a^{n-r} b^r + \ldots\ldots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$

### 2 TRIGONOMETRY

Identities

$\sin^2 A + \cos^2 A = 1$

$\sec^2 A = 1 + \tan^2 A$

$\operatorname{cosec}^2 A = 1 + \cot^2 A$

Formulae for $\Delta ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2} bc \sin A$$

**1** Find the equation of a line parallel to the line $3x - 4y = 12$ passing through the point $(-2, 5)$. [4]

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**2** Solve the simultaneous equations

$x - 2y = 1$,

$y^2 - 3xy + 8 = 0$. [5]

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**3** Given that the line $y - 2x = m$ is a tangent to the curve $x^2 + xy = -12$, find the possible values of $m$. [4]

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**4** Functions $f$ and $g$ are defined by $f: x \longrightarrow \frac{3}{2x+1}, x \neq -\frac{1}{2}$

and $g: x \longrightarrow 2x + 3$.

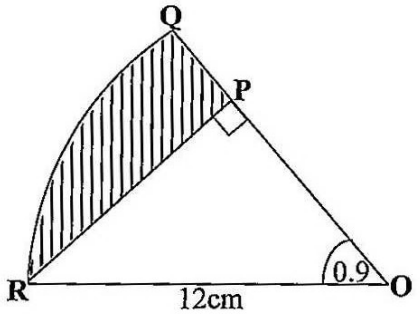
Find

**(a)** the value of $t$, given that $fg(t) = \frac{2}{5}$, [3]

**(b)** the values of $x$ for which $g(x) = f(x)$. [3]

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**5** In the diagram below, ORQ is a sector of a circle with centre O and radius 12cm. RP is perpendicular to OQ at P and angle POR = 0.9 radians.




Find

**(a)** the perimeter of the shaded region, [3]

**(b)** the area of the shaded region. [2]

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**6** Prove the identity $\frac{1-\cos^2\theta}{\sec^2\theta-1} \equiv 1-\sin^2\theta$. [4]

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**7** **(a)** Given that $540x^6$ is the middle term in the expansion of $(a+x^n)^6$ where $a$ is a real number, find the value of $a$ and of $n$. [4]

**(b)** Find the term in $x^3$ in the expansion of $(1+x+x^2)(1-x)^8$. [5]

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**8** **(a)** A curve has the equation $y = \frac{-8}{2x-1}$. Find the value of $x$, for which $\frac{dy}{dx} = 1, x > 0$. [4]

**(b)** Find the equation of the normal to the curve $y = \frac{2x+4}{x-2}$ at the point where $x = 4$. [5]

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**9** Find all the angles between 0° and 360° which satisfy the equation $|\sin x - 0.6| = 0.3$. [5]

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**10** **(a)** The coordinates of the points A and B are (−1, 4) and (2, 3) respectively. Find

**(i)** the magnitude of $\overrightarrow{AB}$, [3]

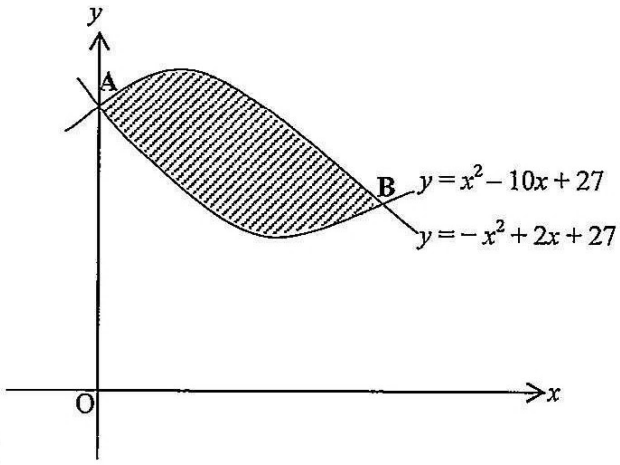
**(ii)** a vector equation of the line AB. [1]

**(b)** Find the angle between the vectors $3\mathbf{i} + 4\mathbf{j}$ and $5\mathbf{i} + 12\mathbf{j}$. [5]

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**11** **(a)** Find the value of $n$ if $\int_0^3 (x^2 + nx)\,dx = 27$. [3]

**(b)** The diagram below shows part of the curve $y = x^2 - 10x + 27$ intersecting the curve $y = -x^2 + 2x + 27$ at A and B.




Find

**(i)** the x-coordinate of point B, [3]

**(ii)** the area of the shaded region. [4]

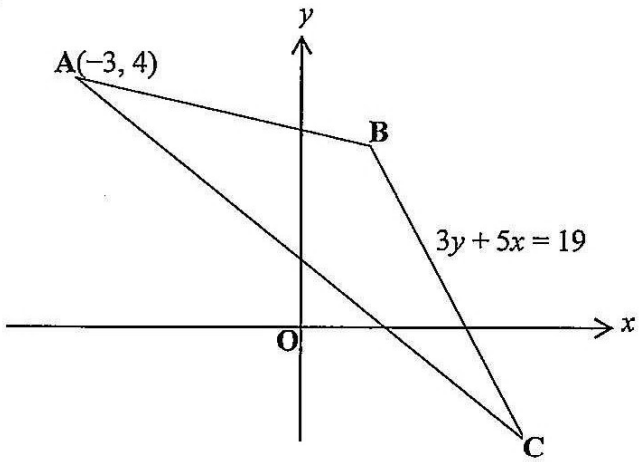
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**Answer only one of the following alternatives.**

**12** **EITHER**

**Solutions to this question by scale drawing will not be accepted.**

In the diagram below, ABC is a triangle. The coordinates of A are (−3, 4) and the equation of BC is $3y + 5x = 19$. The gradients of AB and AC are $-\frac{1}{5}$ and $-\frac{3}{4}$ respectively.



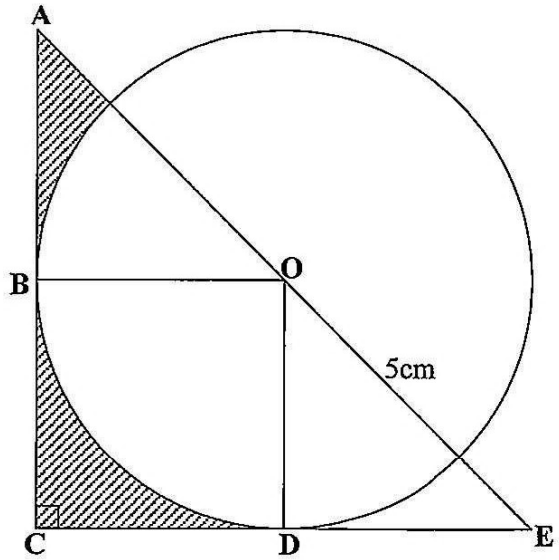

Find

**(a)** the coordinates of B, [4]

**(b)** the magnitude of AC. [6]

**OR**

The diagram shows a circle, centre O, of radius 5cm. ACE is an isosceles triangle such that the adjacent sides AC and CE are tangents to the circle at B and D respectively. Angle ACE is $\frac{\pi}{2}$ radians.




Find

**(a)** the perimeter of the shaded region, [5]

**(b)** the area of the shaded region. [5]



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